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Inventor(s)

Rahn; Jeffrey T. et al.

FREQUENCY DIVISION MULTIPLE ACCESS OPTICAL SUBCARRIERS

Abstract

Methods including a method comprising: transmitting a plurality of optical subcarriers from a first node to a second node in a network; transmitting first control information from the first node to the second node; receiving the plurality of subcarriers and the first control information at the second node; outputting from the second node first data indicative of first information carried by a first group of the plurality of optical subcarriers based on the first control information; transmitting second control information from the first node to the second node; and outputting second data from the second node indicative of second information carried by a second group of the plurality of optical subcarriers based on the second control information.

Inventors: Rahn; Jeffrey T. (San Jose, CA), Wu; Kuang-Tsan (Ottawa, CA), Hand; Steven J. (Los Gatos, CA), Welch; David F. (Atherton, CA)

Applicant: Infinera Corp. (San Jose, CA)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of and claims priority to U.S. application Ser. No. 17/561,805, filed on Dec. 24, 2021, which claims priority to and is a divisional application of U.S. application Ser. No. 16/578,398, filed on Sep. 23, 2019, which claims the benefit of U.S. Provisional Application No. 62/813,151, filed on Mar. 4, 2019, the entire contents of each of which are hereby incorporated herein by reference in their entirety.

BACKGROUND

[0002] In some optical communication systems, multiple optical signals, each having a corresponding wavelength, and each being modulated to carry a different data stream, are multiplexed onto an optical fiber. In such systems, a laser and a modulator may be provided to generate each optical signal. Accordingly, in order to increase the capacity of such systems, additional lasers, modulators and associated circuitry are employed. The cost associated with such systems may, therefore increase as capacity is increased. Accordingly, there is a need for a more cost-effective network requiring fewer components, such as the components described above.

[0003] Moreover, conventional optical communication systems may include high-speed circuitry and components to generate high-speed optical signals at a transmit end of the system. At a receive end, corresponding high-speed circuitry may be provided to detect the incoming data and to forward or distribute such data to lower capacity nodes. Accordingly, there is a further need to reduce costs by supplying high capacity signals to less expensive lower capacity nodes without the need for intermediate high-speed circuitry and components at the receive end of the system.

[0004] In addition, in some conventional optical communication systems, data may be transmitted as a series of frames, each of which includes a payload portion including customer or user data, and a header or overhead portion including operation, administration, and maintenance (“OAM”) information associated with the system. With increasing network complexity, the amount of such control or OAM information has increased, which may limit the amount of transmitted customer data. Therefore, there is also a need to transmit OAM information more efficiently so that more customer data may be transmitted.

SUMMARY

[0005] Consistent the present disclosure, a network or system is provided in which a hub or primary node may communication with a plurality of remote nodes, such as leaf or secondary nodes. The hub node may operate or have a capacity that may be greater than that of the leaf nodes. Accordingly, relatively inexpensive leaf nodes may be deployed that receive data carrying optical signals from, and supply data carrying optical signals to, the hub node. One or more connections may couple each leaf node to the hub node, whereby each connection may include one or more spans or segments of optical fibers, optical amplifiers, optical splitters/combiners, and optical add/drop multiplexer, for example. Consistent with an aspect of the present disclosure, optical subcarriers may be transmitted over such connections, each carrying a data stream. The subcarriers may be generated by a combination of a laser and a modulator, such that multiple lasers and modulators are not required, and costs may be reduced. In addition, the subcarriers may be employed using multiple access techniques, such as frequency division multiplexing access (FDMA), whereby a given group of subcarriers is detected at a corresponding leaf node. Moreover, as the bandwidth or capacity requirements of the leaf nodes change, the number of subcarriers, and thus the amount of data provided to each node may be changed accordingly. In a further example, each subcarrier within a dedicated group of subcarriers may carry OAM or control channel information to a corresponding leaf node, and such information may be used by the leaf node to configure the leaf node to have a desired bandwidth or capacity.

[0006] The foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention, as claimed.

[0007] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several embodiments and together with the description, serve to explain the principles of the invention.

[0008] Other aspects, features and advantages will be apparent from the following detailed description, the accompanying drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate one or more implementations described herein and, together with the description, explain these implementations. In the drawings:

[0010] FIGS. 1 and 2 are block diagrams showing examples of networks consistent with an aspect of the present disclosure;

[0011] FIG. 3 is a block diagram showing an example of a primary node and secondary node, respectively, in accordance with an additional aspect of the present disclosure;

[0012] FIG. 4 is an example of a spectral plot showing optical subcarriers consistent with an aspect of the present disclosure;

[0013] FIG. 5 is a block diagram showing an example of a network consistent with a further aspect of the present disclosure;

[0014] FIG. 6 is an example of a spectral plot showing optical subcarriers consistent with an additional aspect of the present disclosure;

[0015] FIG. 7 is a block diagram showing an example of a network consistent with an additional aspect of the present disclosure;

[0016] FIG. 8 is an example of a spectral plot showing optical subcarriers consistent with a further aspect of the present disclosure;

[0017] FIG. 9 shows an example of a primary node transmitter consistent with the present disclosure;

[0018] FIG. 10A is a block diagram showing an example of a primary node transmitter digital

signal processor (DSP) consistent with a further aspect of the present disclosure;

[0019] FIG. **10B** illustrates a portion of a primary node transmitter DSP in greater detail consistent with an aspect of the present disclosure;

[0020] FIG. **10C** illustrates a portion of a primary node transmitter DSP in greater detail consistent with another aspect of the present disclosure;

[0021] FIG. **11A** shows an example of a receiver circuit in one of secondary nodes consistent with the present disclosure;

[0022] FIG. **11B** shows bandwidths associated with a secondary node being centered about local oscillator frequencies;

[0023] FIG. **12** shows an example of a secondary node receiver DSP consistent with the present disclosure;

[0024] FIG. **13A** shows an example of a secondary node transmitter consistent with an aspect of the present disclosure;

[0025] FIGS. **13A** to **13J** are examples of spectral plots of subcarriers output from secondary nodes consistent with an additional aspect of the present disclosure;

[0026] FIG. **13K** is an example of a spectral plot showing combined subcarriers supplied to a primary node receiver;

[0027] FIG. **14** shows an example of a primary node receiver in accordance with an aspect of the present disclosure;

[0028] FIG. **15** shows an example of a guard band consistent with an additional aspect of the present disclosure;

[0029] FIG. **16** shows an example of a mesh network configuration consistent with a further aspect of the present disclosure;

[0030] FIGS. **17**, **18A-18C**, and **19** show diagrams whereby subcarrier data allocation may change over time consistent with an additional aspect of the present disclosure;

[0031] FIG. **20** is a flow chart of a method for determining time of flight and data transmission timing from consistent with an aspect of the present disclosure;

[0032] FIG. **21** is a flow chart of a method for scheduling;

[0033] FIG. **22** is a plot showing examples of time hub request and leaf response times, as well as corresponding time of flight (TOF) times;

[0034] FIG. **23** shows a table showing TOF times;

[0035] FIG. **24** shows example of timing charts;

[0036] FIG. **25** shows an example of a transceiver.

DESCRIPTION OF THE EMBODIMENTS

[0037] Reference will now be made in detail to the present embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings. In general, the same reference numbers will be used throughout the drawings to refer to the same or like parts. The terms “hub,” “hub node,” and “primary node” are used interchangeably herein. In addition, the terms “leaf,” “leaf node,” and “secondary node” are used interchangeably herein.

[0038] FIG. **1** illustrates an example of an aggregation network **100** consistent with the present disclosure in which primary node **110** may communicate with multiple secondary nodes **112-j** to **112-m**, which sometimes may be referred to individually or collectively as secondary node(s) **112**. Secondary nodes **112**, in one example, are remote from primary node **110**. Primary node **110** may transmit optical subcarriers, described in greater detail below, in a downstream direction onto an optical communication path **111**, which, like each of optical communication paths **113-j** to **113-m**, may include one or more segments of optical fiber, as well as one or more optical amplifiers, reconfigurable add-drop multiplexers (ROADMs) or other optical fiber communication equipment. Splitter **114** may be coupled to an end of optical communication path **111** to receive the optical subcarriers and provide a power split portion of each subcarrier to a corresponding one of secondary nodes **112-j** to **112-m** via a respective one of optical communication paths **113-j** to **113-**

m.

[0039] As further shown in FIG. 1, primary node **110** has a data capacity to receive n Gbit/s of data (e.g., a data stream) for transmission to secondary node **112**. Each secondary node **112** may receive and output to a user or customer a portion of the data input to primary node **110**. In this example, secondary nodes **112-j**, **112-k**, **112-l**, and **112-m** output j Gbit/s, k Gbit/s, l Gbit/s, and m Gbit/s of data (data streams), respectively, whereby the sum of the j , k , l , and m may equal n (where j , k , l , m , and n are positive numbers).

[0040] FIG. 2 show transmission of additional subcarriers in an upstream direction from secondary nodes **112-j** to **112-m** to primary node **110**. As further shown in FIG. 2, each of secondary nodes **112-j** to **112-m** may transmit a corresponding group of subcarriers or one subcarrier to optical combiner **116** via a respective one of optical communication paths **115-l** to **115-m**. Optical combiner **116** may, in turn, combine the received optical subcarriers from secondary nodes **112-j** to **112-m** onto optical communication path **117**. Optical communication paths **115-l** to **115-m** and **117** may have a similar construction as optical communication paths **111** and **112-l** to **112-m**.

[0041] As further shown in FIG. 2, each of secondary nodes **112-j** to **112-m** receives a respective data stream having a corresponding data rate of j Gbit/s, k Gbit/s, l Gbit/s, and m Gbit/s. At primary node **110**, data contained in these streams may be output such that the aggregate data supplied by primary node **110** is n Gbit/s, such that, as noted above, n may equal the sum of j , k , l , and m .

[0042] In another example, subcarriers may be transmitted in both an upstream and downstream direction over the same optical communication path. In particular, selected subcarriers may be transmitted in the downstream direction from primary node **110** to secondary nodes **112**, and other subcarriers may be transmitted in the upstream direction from secondary nodes **112** to primary node **110**.

[0043] In some implementations, network **100** may include additional primary and/or secondary nodes and optical communication paths, fewer primary and/or secondary nodes and optical communication paths, or may have a configuration different from that described above. For example, network **100** may have a mesh configuration or a point-to-point configuration.

[0044] FIG. 3 illustrates primary node **110** in greater detail. Primary node **110** may include a transmitter **202** that supplies a downstream modulated optical signal including subcarriers, and a receiver **204** that may receive upstream subcarriers carrying data originating from the secondary nodes, such as nodes **112-j** to **112-m**.

[0045] FIG. 3 further shows a block diagram of one of secondary nodes **112**, which may include a receiver circuit **302** that receives one or more downstream transmitted subcarriers, and a transmitter circuit **304** that transmits one or more subcarriers in the upstream direction.

[0046] FIG. 4 illustrates an example of a transmission spectrum that can accommodate twenty subcarriers (SC0 to SC19) that may be output from primary node transmitter **202**. Each of subcarriers SC0 to SC19 has a corresponding one of frequencies f_0 to f_{19} . Subcarriers SC0 to SC19, in one example, are Nyquist subcarriers, which are a group of optical signals, each carrying data, wherein (i) the spectrum of each such optical signal within the group is sufficiently non-overlapping such that the optical signals remain distinguishable from each other in the frequency domain, and (ii) such group of optical signals is generated by modulation of light from a single laser. In general, each subcarrier may have an optical spectral bandwidth that is at least equal to the Nyquist frequency, as determined by the baud rate of such subcarrier.

[0047] As noted above, each of secondary nodes **112** may include less expensive components than the components included in primary node **110**. Accordingly, the bandwidth or the data capacity of the secondary nodes **112** may be less than that associated with primary node **110**, such that the capacity associated with each secondary node **112** is less than that of primary node **110**.

[0048] For example, as further shown in FIG. 4, primary node **110** may have a bandwidth BW-P, such that the data carried by each of subcarriers SC1 to SC20 may be processed, recovered, and output either from transmitter **202** or received from receiver **204**. On the other hand, each of

secondary nodes **112-j** to **112-m** may have a respective one of bandwidths BW to BW_m, such that each secondary node has a data processing capacity or is capable of processing and outputting data carried by up to nine subcarriers, in this example.

[0049] As noted above, in order to reduce network costs, less expensive components, such as optical components and certain electrical components, may, in certain examples, be capable of processing signal over of a limited frequency range or bandwidth that is less than the range of signal frequencies that may be accommodated by the optical and electrical components in primary node **110**. For example, electrical components, such as digital-to-analog (DACs), analog-to-digital converters (ADCs), and digital signal processors (DSPs), and optical components, such as modulators, in secondary nodes **112** may have an associated bandwidth that is less than corresponding, albeit more expensive, components in primary node **110**.

[0050] Example bandwidths of each of secondary nodes **112** are further shown in FIG. 4. Namely, bandwidth BW associated with secondary node **112-j** extends over or encompasses a range including frequencies f₀ to f₈ of subcarriers SC₀ to SC₈, respectively; bandwidth BW_k associated with secondary node **112-k** extends over or encompasses a range including frequencies f₅ to f₁₃ of subcarriers SC₅ to SC₁₃, respectively; bandwidth BW_l associated with secondary node **112-l** extends over or encompasses a range including frequencies f₁₀ to f₁₈ of subcarriers SC₁₀ to SC₁₈, respectively; and bandwidth BW_m associated with secondary node **112-m** extends over or encompasses a range including frequencies f₁₁ to f₁₉ of subcarriers SC₁₁ to SC₁₉, respectively. On the other hand, the bandwidth of primary node **110**, BW-P, encompasses the entire range of frequencies f₀-f₁₉ of subcarriers SC₀ to SC₁₉.

[0051] As further shown in FIG. 4, certain subcarriers have frequencies that fall within multiple bandwidths. For example, subcarriers SC₅ and SC₆ have frequencies that fall within bandwidth BW and bandwidth BW_k. The data carried by such subcarriers may, therefore, be detected and selectively output from either secondary node **112-j** or **112-k**. For example, if a customer requires that more data be received and output from secondary node **112-k** and less data be output from secondary node **112-j**, nodes **112-j** and **112-k** may be controlled or dynamically configured such that the data carried by subcarriers SC₅ and SC₆ may be assigned to and output from secondary node **112-k**, but not secondary node **112-j**. Accordingly, as described in greater detail below, the data output from each node may be adapted to customer requirements that vary over time.

[0052] As further shown in FIG. 4, certain subcarriers, such as subcarriers SC₂, SC₇, SC₁₂, and SC₁₇, may be designated or dedicated to carry information related to a parameter or characteristic associated with one or more of secondary nodes **112**. For example, such parameters may correspond to an amount of data, data rate, or capacity to be output by one or more secondary nodes. In particular, such subcarriers may carry information, for example, to configure or adjust the amount of data, capacity or data rate of data output from secondary nodes **112-j** to **112-m**, respectively, as noted above. In a further example, each of these subcarriers may carry user or customer data (also referred to as client data) in addition to control information. In the example shown in FIG. 4, only subcarriers SC₂, SC₇, SC₁₂, and SC₁₇ are transmitted.

[0053] In a further example, subcarriers SC₂, SC₇, SC₁₂, and SC₁₇ are modulated to carry control or OAM information and related data corresponding to parameters associated therewith, such as the capacity and status of nodes **112**. In an additional example, subcarrier SC₂ is modulated carry such control and parameter information associated with node **112-j**, subcarrier SC₇ is modulated to carry such control and parameter information associated with node **112-k**, subcarrier SC₁₂ is modulated to carry such control and parameter information associated with node **112-l** and subcarrier SC₁₇ is modulated to carry such control and parameter information associated with node **112-m**. In a further example, such SCs are modulated to carry information related to a parameter associated the timing and scheduling of data transmission from the nodes **112** to primary node **110**, as described in detail below with respect to FIGS. 19-24.

[0054] FIG. 5 illustrates an example in which each of secondary nodes **112-j** to **112-m** outputs

equal amounts of user data originating from primary node **110**, and FIG. 6 illustrates how subcarriers SC0 to SC19 may be assigned or associated with each secondary node **112** to meet the data allocation shown in FIG. 5. In addition, in the example shown in FIGS. 5 and 6, the aggregate data input to primary node **110** may be 20 Gbit/s ($n=20$), and each of secondary nodes **112** outputs an equal portion of such data, i.e., 5 Gbit/s, whereby $j=k=l=m=5$. Put another way, based on control information received by node **112-j**, for example, data associat

[0055] Here, subcarriers SC0, SC1, SC3, and SC4 may be assigned to secondary node **112-j** (subcarrier SC2 further being assigned to and carrying control channel information for secondary node **112-j**); subcarriers SC5, SC6, SC8, and SC9 may assigned to secondary node **112-k** (subcarrier SC7 further being assigned to and carrying control channel information for secondary node **112-k**); subcarriers SC10, SC11, SC13, and SC14 may assigned to secondary node **112-l** (subcarrier SC12 further being assigned to and carrying control channel information for secondary node **112-l**); and subcarriers SC15, SC16, SC18, and SC19 may assigned to secondary node **112-m** (subcarrier SC17 further being assigned to and carrying control channel information for secondary node **112-m**). Each subcarrier, in this example, may have an associated data rate of 1 Gbit/s.

[0056] Put another way, based on control information received by node **112-j** carried by subcarrier SC2, for example, a first portion of data input to DSP 902 described below and associated with subcarriers SC0, SC1, SC3, and SC4 is output from node **112-j**, as well as any user data carried by subcarrier SC2. Further, based on control information received by node **112-k** carried by subcarrier SC7, for example, a second portion of data input to DSP 902 and associated with subcarriers SC5, SC6, SC8, and SC9 is output from node **112-k**, as well as any user data carried by subcarrier SC7.

[0057] FIG. 7 illustrates another example in which the data allocation to secondary nodes **112-j** to **112-m** may be changed based on, for example, changing capacity requirements. For example, at a point later in time than that associated with the subcarrier allocation shown in FIG. 6, node **112-j**, for example, may require more data than node **112-k**. Here, control information carried by subcarriers SC2, SC7, SC12, and SC17. FIG. 8 illustrates how subcarriers SC0 to SC19 may be assigned or associated with each secondary node **112** to satisfy the data allocation shown in FIG. 7. In this example, secondary node **112-j** is allocated the maximum data capacity, i.e., data carried by seven subcarriers (SC0, SC1 to SC3-SC6, and SC8) as well as any customer data that may be carried by subcarrier SC2; secondary node **112-k** is allocated $\frac{1}{3}$ (i.e., 3/9) the maximum data capacity, i.e., data carried by three subcarriers (SC9-SC11) as well as any customer data that may be carried by subcarrier SC7; secondary node **112-l** is allocated $\frac{5}{9}$ the maximum data capacity, i.e., data carried by four subcarriers (SC13-SC16) as well as any customer data that may be carried by subcarrier SC12; and secondary node **112-m** is allocated a $\frac{1}{3}$ (i.e., 3/9) the maximum data capacity, i.e., data carried by two subcarriers (SC18 and SC19) as well as any customer data that may be carried by subcarrier SC17.

[0058] That is, in the example described above in connection with FIG. 8, based on additional (third) control information received by node **112-j** carried by subcarrier SC2, for example, a third portion of data input to DSP 902 associated with subcarriers SC0, SC1, SC3-SC6, and SC8 is output from node **112-j**, as well as any user data carried by subcarrier SC2. Also, based on (fourth) control information received by node **112-k** carried by subcarrier SC7, for example, a fourth portion of data input to DSP 902 and associated with subcarriers SC9-SC11, is output from node **112-k**, as well as any user data carried by subcarrier SC7.

[0059] Thus, by communicating control information to secondary nodes the amount of data allocated, e.g., the data rate, for output from each secondary node **112** may be controlled or varied over time based on such control information. In the examples discussed above, the control information may identify the subcarriers associated with each node, and, therefore, the amount of data or data rate allocated to each node based on the number of subcarriers carrying data to be output from such node. Such allocation, as noted above, may be changed dynamically, for example, in accordance with varying data traffic requirements in network **100**, and such allocation

information may be carried, for example, by selected or dedicated subcarriers, as noted above, such as subcarriers SC2, SC7, SC12, and SC17.

[0060] Data allocation and subcarrier transmission are described next with reference to FIGS. 9 and 10A-10C.

[0061] FIG. 9 illustrates transmitter 202 of primary node 110 in greater detail. Transmitter 202 includes a plurality of circuits or switches SW, as well as a transmitter DSP (TX DSP) 902 and a D/A and optics block 901. In this example, twenty switches (SW-0 to SW-19) are shown, although more or fewer switches may be provided than that shown in FIG. 9. Each switch may have, in some instances, two inputs: the first input may receive user data, and the second input may receive control information or signals (CNT). Each switch SW-0 to SW-19 can receive a respective one of control signals SWC-0 to SWC-19 output from control circuit 971, which may include a microprocessor, field programmable gate array (FPGA), or other processor circuit. Based on the received control signal, each switch SW-0 to SW-19 selectively outputs any one of the data streams D-0 to D-19, or a control signal CNT-0 to CNT-19. Control signals CNT can be any combination of configuration bits for control and/or monitoring purposes. For example, control signals CNT may include instructions to one or more of secondary nodes 112 to change the data output from such secondary nodes 112, such as by identifying the subcarriers associated with such data. In another example, the control signals may include a series of known bits used in secondary nodes 112 to “train” the receiver to detect and process such bits so that the receiver can further process subsequent bits. In a further example, the control channel CNT includes information that may be used by the polarization mode dispersion (PMD) equalizer circuits 1125 discussed below to correct for errors resulting from polarization rotations of the X and Y components of one or more subcarriers (SC). In a further example, control information CNT is used to restore or correct phase differences between laser transmit-side laser 908 and a local oscillator laser 1110 in each of the secondary nodes 112. Such detected phase differences may be referred to as cycle slips. In a further example, control information CNT may be used to recover, synchronize, or correct timing differences between clocks provided in the primary (110) and secondary nodes 112.

[0062] In another, example, one or more of switches SW may be omitted, and control signals CNT may be supplied directly to DSP 902. Moreover, each input to DSP 902, such as the inputs to FEC encoders 1002 described below (see FIG. 10A), receives, in another example, a combination of control information described above as well as user data.

[0063] In a further example, control signal CNT includes information related to the number of subcarriers that may be output from each of secondary nodes 112. Such selective transmission of subcarriers is described with reference to FIGS. 10A-10C. Although such description is in connection with primary node DSP 902, similar circuitry may be included in the secondary node TX DSP 1302 (FIG. 13A) to adjust or control the number of subcarriers output therefrom.

[0064] Based on the outputs of switches SW-0 to SW-19, DSP 902 may supply a plurality of outputs to D/A and optics block 901 including digital-to-analog conversion (DAC) circuits 904-1 to 904-4, which convert digital signal received from DSP 902 into corresponding analog signals. D/A and optics block 901 also includes driver circuits 906-1 to 906-2 that receive the analog signals from DACs 904-1 to 904-4 and adjust the voltages or other characteristics thereof to provide drive signals to a corresponding one of modulators 910-1 to 910-4.

[0065] D/A and optics block 901 further includes modulators 910-1 to 910-4, each of which may be, for example, a Mach-Zehnder modulator (MZM) that modulates the phase and/or amplitude of the light output from laser 908. As further shown in FIG. 9, light output from laser 908, also included in block 901, is split such that a first portion of the light is supplied to a first MZM pairing, including MZMs 910-1 and 910-2, and a second portion of the light is supplied to a second MZM pairing, including MZMs 910-3 and 910-4. The first portion of the light is split further into third and fourth portions, such that the third portion is modulated by MZM 910-1 to provide an in-phase (I) component of an X (or TE) polarization component of a modulated optical signal, and the

fourth portion is modulated by MZM **910-2** and fed to phase shifter **912-1** to shift the phase of such light by 90 degrees in order to provide a quadrature (Q) component of the X polarization component of the modulated optical signal. Similarly, the second portion of the light is further split into fifth and sixth portions, such that the fifth portion is modulated by MZM **910-3** to provide an I component of a Y (or TM) polarization component of the modulated optical signal, and the sixth portion is modulated by MZM **910-4** and fed to phase shifter **912-2** to shift the phase of such light by 90 degrees to provide a Q component of the Y polarization component of the modulated optical signal.

[0066] The optical outputs of MZMs **910-1** and **910-2** are combined to provide an X polarized optical signal including I and Q components and are fed to a polarization beam combiner (PBC) **914** provided in block **901**. In addition, the outputs of MZMs **910-3** and **910-4** are combined to provide an optical signal that is fed to polarization rotator **913**, further provided in block **901**, that rotates the polarization of such optical signal to provide a modulated optical signal having a Y (or TM) polarization. The Y polarized modulated optical signal also is provided to PBC **914**, which combines the X and Y polarized modulated optical signals to provide a polarization multiplexed (“dual-pol”) modulated optical signal onto optical fiber **916**, for example, which may be included as a segment of optical fiber in optical communication path **111**.

[0067] The polarization multiplexed optical signal output from D/A and optics block **401** includes subcarriers SC0-SC19 noted above, such that each subcarrier has X and Y polarization components and I and Q components. Moreover, each subcarrier SC0 to SC19 may be associated with or corresponds to a respective one of the outputs of switches SW-0 to SW-19. In one example, switches SW2, SW7, SW12, and SW17 may supply control information carried by a respective one of control signals CNT-2, CNT-7, CNT-12, and CNT-17 to DSP **902**. Based on such control signals, DSP **902** provides outputs that result in optical subcarriers SC2, SC7, SC12, and SC17 carrying data indicative of the control information carried by CNT-2, CNT-7, CNT-12, and CNT-17, respectively, as shown in FIGS. 4, 6, and 8. In addition, remaining subcarriers SC0, SC1, SC3 to SC6, SC8 to SC11, SC13 to SC16, and SC18 to SC20 carry information indicative of a respective one of data streams D-0, D-1, D-3-D-6, D-8 to D-11, D-13 to D-16, and D-18 to D-20 output from a corresponding one of switches SW0, SW1, SW3 to SW-6, SW-8 to SW11, SW13 to SW16, and SW18 to SW20, as further shown in FIGS. 4, 6, and 8.

[0068] FIG. 10A shows an example of TX DSP **902** in greater detail. TX DSP **902** may include FEC encoders **1002-0** to **1002-19**, each of which may receive a respective one of a plurality of the outputs from switches SW0 to SW19. FEC encoders **1002-0** to **1002-19** carry out forward error correction coding on a corresponding one of the switch outputs, such as, by adding parity bits to the received data. FEC encoders **1002-0** to **1002-19** may also provide timing skew between the subcarriers to correct for skew induced by link between nodes **110** and **112-j** to **112-m** described above. In addition, FEC encoders **1002-0** to **1002-19** may interleave the received data.

[0069] Each of FEC encoders **1002-0** to **1002-19** provides an output to a corresponding one of a plurality of bits-to-symbol circuits, **1004-0** to **1004-19** (collectively referred to herein as “**1004**”). Each of bits-to-symbol circuits **1004** may map the encoded bits to symbols on a complex plane. For example, bits-to-symbol circuits **1004** may map four bits to a symbol in a dual-polarization QPSK constellation. Each of bits-to-symbol circuits **1004** provides first symbols, having the complex representation $XI+j*XQ$, associated with a respective one of the switch outputs, such as D-0, to DSP portion **1003**. Data indicative of such first symbols is carried by the X polarization component of each subcarrier SC0-SC19.

[0070] Each of bits-to-symbol circuits **1004** further may provide second symbols having the complex representation $YI+j*YQ$, also associated with a corresponding output of switches SW0-SW19. Data indicative of such second symbols, however, is carried by the Y polarization component of each of subcarriers SC-0 to SC-19.

[0071] Such mapping, as carried by about circuit **1004-0** to **1004-19** define, in one example, a

particular modulation format for each subcarrier. That is, such circuit may define a mapping for all the optical subcarrier that is indicative of a binary phase shift keying (BPSK) modulation format, a quadrature phase shift keying (QPSK) modulation format, or an m-quadrature amplitude modulation (QAM, where m is a positive integer, e.g., 4, 8, 16, or 64) format. In another example, one or more of the optical subcarriers may have a modulation format that is different than the modulation format of other optical subcarriers. That is, one of the optical subcarriers have a QPSK modulation format and another optical subcarrier has a different modulation format, such as 8-QAM or 16-QAM. In another example, one of the optical subcarriers has an 8-QAM modulation format and another optical subcarrier has a 16 QAM modulation format. Accordingly, although all the optical subcarriers may carry data at the same data and or baud rate, consistent with an aspect of the present disclosure one or more of the optical subcarriers may carry data at a different data or baud rate than one or more of the other optical subcarriers. Moreover, modulation formats, baud rates and data rates may be changed over time depending on capacity requirements, for example. Adjusting such parameters may be achieved, for example, by applying appropriate signals to mappers **1004** based on control information or data described herein and the communication of such data as further disclosed herein between hub and leaf nodes.

[0072] As further shown in FIG. **10A**, each of the first symbols output from each of bits-to-symbol circuits **1004** is supplied to a respective one of first overlap and save buffers **1005-0** to **1005-19** (collectively referred to herein as overlap and save buffers **1005**) that may buffer 256 symbols, for example. Each of overlap and save buffers **1005** may receive 128 of the first symbols or another number of such symbols at a time from a corresponding one of bits to symbol circuits **1004**. Thus, overlap and save buffers **1005** may combine 128 new symbols from bits to symbol circuits **1005**, with the previous 128 symbols received from bits to symbol circuits **1005**.

[0073] Each overlap and save buffer **1005** supplies an output, which is in the time domain, to a corresponding one of fast Fourier Transform (FFT) circuits **1006-0** to **1006-19** (collectively referred to as “FFTs **1006**”). In one example, the output includes 256 symbols or another number of symbols. Each of FFTs **1006** converts the received symbols to the frequency domain using or based on, for example, a fast Fourier transform. Each of FFTs **1006** may provide the frequency domain data to bins and switches blocks **1021-0** to **1021-19**. As discussed in greater detail below, bins and switches blocks **1021** include, for example, memories or registers, also referred to as frequency bins (FB) or points, that store frequency components associated with each subcarrier SC.

[0074] Selected frequency bins FB are shown in FIG. **10B**. Groups of such frequency bins FB are associated with give subcarriers. Accordingly, for example, a first group of frequency bins, FB**0-0** to FB**0-n**, is associated with SC**0** and a second group of frequency bins FB**19-0** to FB**19-n** with SC**19** (where n is a positive integer). As further shown in FIG. **10B**, each of frequency bins FB is further coupled to a respective one of switches SW. For example, each of frequency bins FB**0-0** to FB**0-n** is coupled to a respective one of switches SW**0-0** to SW**0-n**, and each of FB**19-0** to FB**19-n** is coupled to a respective one of switches or switch circuits SW**19-0** to SW**19-n**.

[0075] Each switch SW selectively supplies either frequency domain data output from one of FFT circuits **1006-0** to **1006-19** or a predetermined value, such as 0. In order to block or eliminate transmission of a particular subcarrier, the switches SW associated with the group of frequency bins FB associated with that subcarrier are configured to supply the zero value to corresponding frequency bins. Accordingly, for example, in order to block subcarrier SC**0**, switches SW**0-0'** to SW**0-n'** supply zero (0) values to a respective one of frequency bins FB**0-0** to FB**0-n**. Further processing, as described below, of the zero (0) values by replicator components **1007** as well as other components and circuits in DSP **902** result in drive signals supplied to modulators **910**, such that subcarrier SC**0** is omitted from the optical output from the modulators.

[0076] On the other hand, switches SW may be configured to supply the outputs of FFTs **1006**, i.e., frequency domain data FD, to corresponding frequency bins FB. Further processing of the contents of frequency bins FB by replicator components **1007** and other circuits in DSP **902** result in drive

signals supplied to modulators **910**, whereby, based on such drive signals, optical subcarriers are generated that correspond to the frequency bin groupings associated with that subcarrier.

[0077] In the example discussed above, switches **SW0-0'** to **SW0-n'** supply frequency domain data **FD0-0** to **FD-n** from FFT **1006-0** to a respective one of switches **SW0-0** to **SW0-n**. These switches, in turn, supply the frequency domain data to a respective one of frequency bins **FB0-0** to **FB0-n** for further processing, as described in greater detail below.

[0078] Each of replicator components or circuits **1007-0** to **1007-19** may replicate the contents of the frequency bins **FB** and store such contents (e.g., for T/2 based filtering of the subcarrier) in a respective one of the plurality of replicator components. Such replication may increase the sample rate. In addition, replicator components or circuits **1007-0** to **1007-19** may arrange or align the contents of the frequency bins to fall within the bandwidths associated with pulse shaped filter circuits **1008-0** to **1008-19** described below.

[0079] Each of pulse shape filter circuits **1008-0** to **1008-19** may apply a pulse shaping filter to the data stored in the 512 frequency bins of a respective one of the plurality of replicator components or circuits **1007-0** to **1007-19** to thereby provide a respective one of a plurality of filtered outputs, which are multiplexed and subject to an inverse FFT, as described below. Pulse shape filter circuits **1008-1** to **1008-19** calculate the transitions between the symbols and the desired subcarrier spectrum so that the subcarriers can be packed together spectrally for transmission, e.g., with a close frequency separation. Pulse shape filter circuits **1008-0** to **1008-19** also may be used to introduce timing skew between the subcarriers to correct for timing skew induced by links between nodes shown in FIG. 1, for example. Multiplexer component **1009**, which may include a multiplexer circuit or memory, may receive the filtered outputs from pulse shape filter circuits **1008-0** to **1008-19**, and multiplex or combine such outputs together to form an element vector.

[0080] Next, IFFT circuit or component **1010-1** may receive the element vector and provide a corresponding time domain signal or data based on an inverse fast Fourier transform (IFFT). In one example, the time domain signal may have a rate of 64 GSample/s. Take last buffer or memory circuit **1011-1**, for example, may select the last 1024 samples, or another number of samples, from an output of IFFT component or circuit **1010-1** and supply the samples to DACs **904-1** and **904-2** (see FIG. 9) at 64 GSample/s, for example. As noted above, DAC **904-1** is associated with the in-phase (I) component of the X pol signal, and DAC **904-2** is associated with the quadrature (Q) component of the Y pol signal. Accordingly, consistent with the complex representation $XI + jXQ$, DAC **904-1** receives values associated with XI and DAC **904-2** receives values associated with jXQ . As indicated by FIG. 9, based on these inputs, DACs **904-1** and **904-2** provide analog outputs to MZMD **906-1** and MZMD **906-2**, respectively, as discussed above.

[0081] As further shown in FIG. 10A, each of bits-to-symbol circuits **1004-0** to **1004-19** outputs a corresponding one of symbols indicative of data carried by the Y polarization component of the polarization multiplexed modulated optical signal output on fiber **916**. As further noted above, these symbols may have the complex representation $YI + j*YQ$. Each such symbol may be processed by a respective one of overlap and save buffers **1015-0** to **1015-19**, a respective one of FFT circuits **1016-0** to **1016-19**, a respective one of replicator components or circuits **1017-0** to **517-19**, pulse shape filter circuits **1018-0** to **1018-19**, multiplexer or memory **1019**, IFFT **1010-2**, and take last buffer or memory circuit **1011-2**, to provide processed symbols having the representation $YI + j*YQ$ in a manner similar to or the same as that discussed above in generating processed symbols $XI + j*XQ$ output from take last circuit **1011-1**. In addition, symbol components YI and YQ are provided to DACs **904-3** and **904-4** (FIG. 9), respectively. Based on these inputs, DACs **904-3** and **904-4** provide analog outputs to MZMD **906-3** and MZMD **906-4**, respectively, as discussed above.

[0082] While FIG. 10A shows DSP **902** as including a particular number and arrangement of functional components, in some implementations, DSP **902** may include additional functional components, fewer functional components, different functional components, or differently arranged

functional components. In addition, typically the number of overlap and save buffers, FFTs, replicator circuits, and pulse shape filters associated with the X component may be equal to the number of switch outputs, and the number of such circuits associated with the Y component may also be equal to the number of switch outputs. However, in other examples, the number of switch outputs may be different from the number of these circuits.

[0083] As noted above, based on the outputs of MZMDs **906-1** to **906-4**, a plurality of optical subcarriers SC0 to SC19 may be output onto optical fiber **916** (FIG. 9), which is coupled to the primary node **110**.

[0084] Consistent with an aspect of the present disclosure, the number of subcarriers transmitted from primary node **110** to secondary nodes **112** may vary over time based, for example, on capacity requirements at the primary node and the secondary nodes. For example, if less downstream capacity is required initially at one or more of the secondary nodes, transmitter **202** in primary node **110** may be configured to output fewer optical subcarriers. On the other hand, if further capacity is required later, transmitter **202** may provide more optical subcarriers.

[0085] In addition, if based on changing capacity requirements, a particular secondary node **112** needs to be adjusted, for example, the output capacity of such secondary node may be increased or decreased by, in a corresponding manner, increasing or decreasing the number of optical subcarriers output from the secondary node.

[0086] As noted above, by storing and subsequently processing zeros (0s) or other predetermined values in frequency bin FB groupings associated with a given subcarrier SC, that subcarrier may be removed or eliminated. To add or reinstate such subcarrier, frequency domain data output from the FFTs **1006** may be stored in frequency bins FB and subsequently processed to provide the corresponding subcarrier. Thus, subcarriers may be selectively added or removed from the optical outputs of primary node transmitter **202** and secondary node transmitter **304**, such that the number of subcarriers output from such transmitters may be varied, as desired.

[0087] In the above example, zeros (0s) or other predetermined values are stored in selected frequency bins FBs to prevent transmission of a particular subcarrier SC. Such zeroes or values may, instead, be provided, for example, in a manner similar to that described above, at the outputs of corresponding replicator components **1007** or stored in corresponding locations in memory or multiplexer **1009**. Alternatively, the zeroes or values noted above may be provided, for example, in a manner similar to that described above, at corresponding outputs of pulse shape filters **1008**.

[0088] In a further example, a corresponding one of pulse shape filters **1008-1** to **1008-19** may selectively generate zeroes or predetermined values that, when further processed, also cause one or more subcarriers SC to be omitted from the output of either primary node transmitter **202** or secondary node transmitter **304**. In particular, as shown in FIG. 10C, pulse shape filters **1008-0** to **1008-19** are shown as including groups of multiplier circuits M0-0 to M0- n . . . M19-0 to M19- n (also individually or collectively referred to as M). Each multiplier circuit M constitutes part of a corresponding butterfly filter. In addition, each multiplier circuit grouping is associated with a corresponding one of subcarriers SC.

[0089] Each multiplier circuit M receives a corresponding one of output groupings RD0-0 to RD0- n . . . RD19-0 to RD19- n from replicator components **1007**. In order to remove or eliminate one of subcarriers SC, multiplier circuits M receiving the outputs within a particular grouping associated with that subcarrier multiply such outputs by zero (0), such that each multiplier M within that group generates a product equal to zero (0). The zero products then are subject to further processing similar to that described above to provide drive signals to modulators **910** that result in a corresponding subcarrier SC being omitted from the output of the transmitter (either transmitter **202** or **304**).

[0090] On the other hand, in order to provide a subcarrier SC, each of the multiplier circuits M within a particular grouping may multiply a corresponding one of replicator outputs RD by a respective one of coefficients C0-0 to C0- n . . . C19-0 to C19- n , which results in at least some non-

zero products being output. Based on the products output from the corresponding multiplier grouping, drive signals are provided to modulators **910** to output the desired subcarrier SC from the transmitter (either transmitter **202** or **304**).

[0091] Accordingly, for example, in order to block or eliminate subcarrier SC0, each of multiplier circuits M0-0 to M0-n (associated with subcarrier SC0) multiplies a respective one of replicator outputs RD0-0 to RD0-n by zero (0). Each such multiplier circuit, therefore, provides a product equal to zero, which is further processed, as noted above, such that resulting drive signals cause modulators **910** to provide an optical output without SC0. In order to reinstate SC0, multiplier circuits M0-0 to M0-n multiply a corresponding one of appropriate coefficients C0-0 to C0-n by a respective one of replicator outputs RD0-0 to RD0-n to provide products, at least some of which are non-zero. Based on these products, as noted above, modulator drive signals are generated that result in subcarrier SC0 being output.

[0092] The above examples are described in connection with generating or removing the X component of a subcarrier SC. The processes and circuitry described above is employed or included in DSP **902** and optical circuitry used to generate the Y component of the subcarrier to be blocked. For example, switches and bins circuit blocks **1022-0** to **1022-19**, have a similar structure and operate in a similar manner as switches and bins circuit blocks **1021** described above to provide zeroes or frequency domain data as the case may be to selectively block the Y component of one or more subcarriers SC. Alternatively, multiplier circuits, like those described above in connection with FIG. **10C** may be provided to supply zero products output from selected pulse shape filters **1018** in order to block the Y component of a particular subcarrier or, if non-zero coefficients are provided to the multiplier circuits instead, generate the subcarrier.

[0093] Thus, the above examples illustrate mechanisms by which subcarriers SC may be selectively blocked from or added to the output of transmitter **202**. Since, as discussed below, DSPs and optical circuitry provided in secondary node transmitters **304** are similar to that of primary node transmitter **202**, the processes and circuitry described above is provided, for example, in the secondary node transmitters **304** to selectively add and remove subcarriers SC' from the outputs of the secondary node transmitters, as described in connection with FIGS. **13B-13K**. Moreover, consistent with the present disclosure, the circuitry described above in connection with FIGS. **10B** and/or **10C** maybe configured so that a first number of optical subcarriers are output from the transmitter (in either the primary node **110** or the secondary node **112**) during a first period of time based on initial capacity requirements. Later, during a second period of time, a second number of optical subcarriers can be output from the hub and/or leaf transmitters based on capacity requirements different than the first capacity requirements.

[0094] Optical subcarriers SC0 to SC19 may be provided to secondary nodes **112** in FIG. **1**. An example of receiver circuit **302** in one of secondary nodes **112** will be described next with reference to FIG. **11A**.

[0095] As shown in FIG. **11A**, optical receiver **302** may include an Rx optics and A/D block **1100**, which, in conjunction with DSP **1150**, may carry out coherent detection. Block **1100** may include a polarization splitter (PBS) **1105** with first (**1105-1**) and second (**1105-2**) outputs), a local oscillator (LO) laser **1110**, 90 degree optical hybrids or mixers **1120-1** and **1120-2** (referred to generally as hybrid mixers **1120** and individually as hybrid mixer **1120**), detectors **1130-1** and **1130-2** (referred to generally as detectors **1130** and individually as detector **1130**, each including either a single photodiode or balanced photodiode), AC coupling capacitors **1132-1** and **1132-2**, transimpedance amplifiers/automatic gain control circuits TIA/AGC **1134-1** and **1134-2**, ADCs **1140-1** and **1140-2** (referred to generally as ADCs **1140** and individually as ADC **1140**).

[0096] Polarization beam splitter (PBS) **1105** may include a polarization splitter that receives an input polarization multiplexed optical signal including optical subcarriers SC0 to SC19 supplied by optical fiber link **1101**, which may be, for example, an optical fiber segment as part of one of optical communication paths **113-k** to **113-m** noted above. PBS **1105** may split the incoming optical

signal into the two X and Y orthogonal polarization components. The Y component may be supplied to a polarization rotator **1106** that rotates the polarization of the Y component to have the X polarization. Hybrid mixers **1120** may combine the X and rotated Y polarization components with light from local oscillator laser **1110**, which, in one example, is a tunable laser. For example, hybrid mixer **1120-1** may combine a first polarization signal (e.g., the component of the incoming optical signal having a first or X (TE) polarization output from a first PBS port with light from local oscillator **1110**, and hybrid mixer **1120-2** may combine the rotated polarization signal (e.g., the component of the incoming optical signal having a second or Y (TM) polarization output from a second PBS port) with the light from local oscillator **1110**. In one example, polarization rotator **1190** may be provided at the PBS output to rotate Y component polarization to have the X polarization.

[0097] Detectors **1130** may detect mixing products output from the optical hybrids, to form corresponding voltage signals, which are subject to AC coupling by capacitors **1132-1** and **1132-1**, as well as amplification and gain control by TIA/AGCs **1134-1** and **1134-2**. The outputs of TIA/AGCs **1134-1** and **1134-2** and ADCs **1140** may convert the voltage signals to digital samples. For example, two detectors (e.g., photodiodes) **1130-1** may detect the X polarization signals to form the corresponding voltage signals, and a corresponding two ADCs **1140-1** may convert the voltage signals to digital samples for the first polarization signals after amplification, gain control and AC coupling. Similarly, two detectors **1130-2** may detect the rotated Y polarization signals to form the corresponding voltage signals, and a corresponding two ADCs **1140-2** may convert the voltage signals to digital samples for the second polarization signals after amplification, gain control and AC coupling. RX DSP **1150** may process the digital samples associated with the X and Y polarization components to output data associated with one or more subcarriers within a group of subcarriers SC0 to SC19 encompassed by the bandwidth (one of bandwidths BW, BWk, BWl, and BWm) associated with the secondary node housing the particular DSP **1150**. For example, as shown in FIG. 6, subcarriers SC0 to SC8 are within bandwidth BSj, and such subcarriers may be processed by the receiver in secondary node **112-j**. As noted above, however, subcarriers SC5 to SC13 are within bandwidth BWk may be processed by the receiver in secondary node **112-k**. That is, bandwidths BW and BWk overlap, such that subcarriers within the overlapped portions of these bandwidths, namely, subcarriers SC5 to SC8, will be processed by the receivers in both secondary node **112-j** and **112-k**. If the data associated with these subcarriers is intended to be output from secondary node **112-k**, switch circuits, for example, may be provided to selectively output such data at node **112-k** but not from node **112-j**, as discussed in greater detail below.

[0098] While FIG. 11A shows optical receiver **302** as including a particular number and arrangement of components, in some implementations, optical receiver **302** may include additional components, fewer components, different components, or differently arranged components. The number of detectors **1130** and/or ADCs **1140** may be selected to implement an optical receiver **302** that is capable of receiving a polarization multiplexed signal. In some instances, one of the components illustrated in FIG. 11A may carry out a function described herein as being carry out by another one of the components illustrated in FIG. 11A.

[0099] Consistent with the present disclosure, in order to select a particular subcarrier or group of subcarriers at a secondary node **112**, local oscillator **1110** may be tuned to output light having a wavelength or frequency relatively close to the selected subcarrier wavelength(s) to thereby cause a beating between the local oscillator light and the selected subcarrier(s). Such beating will either not occur or will be significantly attenuated for the other non-selected subcarriers so that data carried by the selected subcarrier(s) is detected and processed by DSP **1150**.

[0100] As noted above, each secondary node **112** may have a smaller bandwidth than the bandwidth associated with primary node **110**. The subcarriers encompassed by each secondary node may be determined by the frequency of the local oscillator laser **1110** in the secondary node receiver **302**. For example, as shown in FIG. 11B, bandwidth BW associated with secondary node

112-j may be centered about local oscillator frequency fLO_j , bandwidth BW_k associated with secondary node **112-k** may be centered about local oscillator frequency fLO_k , bandwidth BW_l associated with secondary node **112-l** may be centered about local oscillator frequency fLO_l , and bandwidth BW_m associated with secondary node **112-m** may be centered about local oscillator frequency fLO_m . Accordingly, each bandwidth BW to BW_m may shift depending on the frequency of each secondary node local oscillator laser **1110**. Tuning the local oscillator frequency, for example, by changing the temperature of the local oscillator laser **1110** may result in corresponding shifts in the bandwidth to encompass a different group of subcarriers than were detected prior to such bandwidth shift. The temperature of the local oscillator laser **1110** may be controlled with a thin film heater, for example, provided adjacent the local oscillator laser or to portions of the local oscillator laser such as the mirror sections. Alternatively, the local oscillator laser may be frequency tuned by controlling the current supplied to the laser. The local oscillator laser **1110** may be a semiconductor laser, such as a distributed feedback laser or a distributed Bragg reflector laser. [0101] The maximum bandwidth or number of subcarriers that may be received, detected, and processed by a secondary node receiver **302**, however, may be restricted based on hardware limitations of the various circuit components in receiver **302**, as noted above, and, therefore, may be fixed, in this example. Accordingly, as noted above, the bandwidth associated with each secondary node **112** may be less than bandwidth $BW-P$ associated with primary node **110**. Further, consistent with the present disclosure, the number of secondary nodes may be greater than the number of subcarriers output from primary node **110**. In addition, the number of upstream subcarriers received by primary node **110** may be equal to the number of subcarriers transmitted by primary node **110** in the upstream direction. Alternatively, the number of subcarriers transmitted in the upstream direction collectively by secondary nodes **112** may less than or greater than the number of downstream subcarriers output from the primary node. Further, in another example, one or more of secondary nodes **112** may output a single subcarrier.

[0102] As shown in FIG. **11B** and discussed above, the bandwidths associated with secondary nodes **112** may overlap, such that, as further noted above, certain subcarriers SC may be detected by multiple secondary nodes **112**. If the data associated with such subcarriers SC is intended for one of those secondary nodes, but not the other, switch circuitry, as noted above, may be provided in the secondary nodes to output the data selectively at the intended secondary node but not the others.

[0103] For example, as further shown in FIG. **11A**, switches or circuits $SW-0$ to $SW-8$ may be provided at the output of DSP **1150** to selectively output the data detected from the received subcarriers based on a respective one of control signals $CNT-0$ to $CNT-8$ output from control circuit **1171**, which, like control circuit **971** noted above may include a microprocessor, FPGA, or other processor circuit. Control signals may designate the output of each respective switch. Accordingly, for example, if data carried by predetermined subcarriers is intended to be output at a particular secondary node **112**, switches SW at that secondary node may be configured, based on the received control signals CNT , to supply the desired data, but block data not intended for that node.

[0104] FIG. **12** illustrates exemplary components of receiver digital signal processor (DSP) **1150**. As noted above, analog-to-digital (A/D) circuits **1140-1** and **1140-2** (FIG. **11A**) output digital samples corresponding to the analog inputs supplied thereto. In one example, the samples may be supplied by each A/D circuit at a rate of 64 GSamples/s. The digital samples correspond to symbols carried by the X polarization of the optical subcarriers and may be represented by the complex number $XI+jXQ$. The digital samples may be provided to overlap and save buffer **1205-1**, as shown in FIG. **12**. FFT component or circuit **1210-1** may receive the 2048 vector elements, for example, from the overlap and save buffer **1205-1** and convert the vector elements to the frequency domain using, for example, a fast Fourier transform (FFT). The FFT component **1210-1** may convert the 2048 vector elements to 2048 frequency components, each of which may be stored in a register or

“bin” or other memory, as a result of carrying out the FFT.

[0105] The frequency components then may be demultiplexed by demultiplexer **1211-1**, and groups of such components may be supplied to a respective one of chromatic dispersion equalizer circuits CDEQ **1212-1-0** to **1212-1-8**, each of which may include a finite impulse response (FIR) filter that corrects, offsets or reduces the effects of, or errors associated with, chromatic dispersion of the transmitted optical subcarriers. Each of CDEQ circuits **1212-1-0** to **1212-1-8** supplies an output to a corresponding polarization mode dispersion (PMD) equalizer circuit **1225-0** to **1225-8** (which individually or collectively may be referred to as **1225**).

[0106] Digital samples output from A/D circuits **640-2** associated with Y polarization components of subcarrier SC1 may be processed in a similar manner to that of digital samples output from A/D circuits **1240-1** and associated with the X polarization component of each subcarrier. Namely, overlap and save buffer **1205-2**, FFT **1210-2**, demultiplexer **1211-2**, and CDEQ circuits **1212-2-0** to **1212-2-8** may have a similar structure and operate in a similar fashion as buffer **1205-1**, FFT **1210-1**, demultiplexer **122-1**, and CDEQ circuits **1212-1-0** to **1212-1-8**, respectively. For example, each of CDEQ circuits **1212-2-0** to **1212-8** may include an FIR filter that corrects, offsets, or reduces the effects of, or errors associated with, chromatic dispersion of the transmitted optical subcarriers. In addition, each of CDEQ circuits **1212-2-0** to **1212-2-8** provide an output to a corresponding one of PMDEQ **1225-0** to **1225-8**.

[0107] As further shown in FIG. **12**, the output of one of the CDEQ circuits, such as CDEQ **1212-1-0** may be supplied to clock phase detector circuit **1213** to determine a clock phase or clock timing associated with the received subcarriers. Such phase or timing information or data may be supplied to ADCs **1140-1** and **1140-2** to adjust or control the timing of the digital samples output from ADCs **1140-1** and **1140-2**.

[0108] Each of PMDEQ circuits **1225** may include another FIR filter that corrects, offsets or reduces the effects of, or errors associated with, PMD of the transmitted optical subcarriers. Each of PMDEQ circuits **1225** may supply a first output to a respective one of IFFT components or circuits **1230-0-1** to **1230-8-1** and a second output to a respective one of IFFT components or circuits **1230-0-2** to **1230-8-2**, each of which may convert a 256-element vector, in this example, back to the time domain as 256 samples in accordance with, for example, an inverse fast Fourier transform (IFFT).

[0109] Time domain signals or data output from IFFT **1230-0-1** to **1230-8-1** are supplied to a corresponding one of Xpol carrier phase correction circuits **1240-1-1** to **1240-8-1**, which may apply carrier recovery techniques to compensate for X polarization transmitter (e.g., laser **908**) and receiver (e.g., local oscillator laser **1110**) linewidths. In some implementations, each carrier phase correction circuit **1240-0-1** to **1240-8-1** may compensate or correct for frequency and/or phase differences between the X polarization of the transmit signal and the X polarization of light from the local oscillator **1100** based on an output of Xpol carrier recovery circuit **1240-0-1**, which performs carrier recovery in connection with one of the subcarrier based on the outputs of IFFT **1230-01**. After such X polarization carrier phase correction, the data associated with the X polarization component may be represented as symbols having the complex representation $x_i + j \cdot x_q$ in a constellation, such as a QPSK constellation or a constellation associated with another modulation formation, such as an m-quadrature amplitude modulation (QAM), m being an integer. In some implementations, the taps of the FIR filter included in one or more of PMDEQ circuits **1225** may be updated based on the output of at least one of carrier phase correction circuits **1240-0-1** to **1240-8-1**.

[0110] In a similar manner, time domain signals or data output from IFFT **1230-0-2** to **1230-8-2** are supplied to a corresponding one of Ypol carrier phase correction circuits **1240-0-2** to **1240-8-2**, which may compensate or correct for Y polarization transmitter (e.g., laser **908**) and receiver (e.g., local oscillator laser **1110**) linewidths. In some implementations, each carrier phase correction circuit **1240-0-2** to **1240-8-2** also may correct or compensate for frequency and/or phase

differences between the Y polarization of the transmit signal and the Y polarization of light from the local oscillator **1110**. After such Y polarization carrier phase correction, the data associated with the Y polarization component may be represented as symbols having the complex representation $y_i + j \cdot y_q$ in a constellation, such as a QPSK constellation or a constellation associated with another modulation formation, such as an m-quadrature amplitude modulation (QAM), m being an integer. In some implementations, the output of one of circuits **1240-0-2** to **1240-8-2** may be used to update the taps of the FIR filter included in one or more of PMDEQ circuits **1225** instead of, or in addition to, the output of at least one of the carrier recovery circuits **1240-0-1** to **1240-8-1**.

[0111] As further shown in FIG. **12**, the output of carrier recovery circuits, e.g., carrier recovery circuit **1240-0-1**, also may be supplied to carrier phase correction circuits **1240-1-1** to **1240-8-1** and **1240-0-2** to **1240-8-2**, whereby the phase correction circuits may determine or calculate a corrected carrier phase associated with each of the received subcarriers based on one of the recovered carriers, instead of providing multiple carrier recovery circuits, each of which is associated with a corresponding subcarrier. The equalizer, carrier recovery, and clock recovery may be further enhanced by utilizing the known (training) bits that may be included in control signals CNT, for example by providing an absolute phase reference between the transmitted and local oscillator lasers.

[0112] Each of the symbols-to-bits circuits or components **1245-0-1** to **1245-8-1** may receive the symbols output from a corresponding one of circuits **1240-0-1** to **1240-8-1** and map the symbols back to bits. For example, each of the symbol-to-bits components **1245-0-1** to **1245-8-1** may map one X polarization symbol, in a QPSK or m-QAM constellation, to Z bits, where Z is an integer. For dual-polarization QPSK modulated subcarriers, Z is four. Bits output from each of component **1245-0-1** to **1245-8-1** are provided to a corresponding one of FEC decoder circuits **1260-0** to **1260-8**.

[0113] Y polarization symbols are output from a respective one of circuits **1240-0-2** to **1240-8-2**, each of which has the complex representation $y_i + j \cdot y_q$ associated with data carried by the Y polarization component. Each Y polarization, like the X polarization symbols noted above, may be provided to a corresponding one of bit-to-symbol circuits or components **1245-0-2** to **1245-8-2**, each of which has a similar structure and operates in a similar manner as symbols-to-bits component **1245-0-1** to **1245-8-1**. Each of circuits **1245-0-2** to **1245-8-2** may provide an output to a corresponding one of FEC decoder circuits **1260-0** to **1260-8**.

[0114] Each of FEC decoder circuits **1260** may remove errors in the outputs of symbol- to-bit circuits **1245** using, for example, forward error correction. Such error corrected bits, which may include user data for output from secondary node **112**, may be supplied to a corresponding one of switch circuits SW-**0** to SW-**8**. As noted above, switch circuits SW-**0** to SW-**8** in each secondary node **112** may selectively supply or block data based on whether such data is intended to be output from the secondary node. In addition, if one of the received subcarriers' control information (CNT), such as information identifying switches SW that output data and other switches SW that block data, the control information may be output from one of the switches and, based on such control information, control circuit **1171** in the secondary nodes to generate the control signals CNT.

[0115] Consistent with another aspect of the present disclosure, data may be blocked from output from DSP **1150** without the use of switches SW-**0** to SW-**8**. In one example similar to an example described above, zero (0) or other predetermined values may be stored in frequency bins associated with the blocked data, as well as the subcarrier corresponding to the blocked data. Further processing described above of such zeroes or predetermined data by circuitry in DSP **1150** will result in null or zero data outputs, for example, from a corresponding one of FEC decoders **1260**. Switch circuits provided at the outputs of FFTs **1210-1** and **1210-2**, like switch circuits SW described above in FIG. **10B**, may be provided to selectively insert zeroes or predetermined values for selectively blocking corresponding output data from DSP **1150**. Such switches also may be provided at the output of or within demultiplexers **1211-1** and **1211-2** to selectively supply zero or

predetermined values.

[0116] In another example, zeroes (0s) may be inserted in chromatic dispersion equalizer (CDEQ) circuits **1212** associated with both the X and Y polarization components of each subcarrier. In particular, multiplier circuits (provided in corresponding butterfly filter circuits), like multiplier circuits **M** described above, may selectively multiply the inputs to the CDEQ circuit **1212** by either zero or a desired coefficient. As discussed above in connection with FIG. **10C**, multiplication by a zero generates a zero product. When such zero products are further processed by corresponding circuitry in DSP **1150**, e.g., corresponding IFFTs **1230**, carrier phase correction components **1240**, symbol-to-bits components **1245**, and FEC decoder, a corresponding output of DSP **1150** will also be zero. Accordingly, data associated with a subcarrier SC received by a secondary node receiver **112**, but not intended for output from that receiver, can be blocked.

[0117] If, on the other hand, capacity requirements change and such previously blocked data is to be output from a given secondary node receiver DSP **1150**, appropriately coefficients may be supplied to the multiplier circuits, such that at least some of the inputs thereto are not multiplied by zero. Upon further processing, as noted above, data associated with the inputs to the multiplier circuits and corresponding to a particular subcarrier SC is output from secondary node receiver DSP **1150**.

[0118] While FIG. **12** shows DSP **1150** as including a particular number and arrangement of functional components, in some implementations, DSP **650** may include additional functional components, fewer functional components, different functional components, or differently arranged functional components.

[0119] Upstream transmission from a secondary node **112** to primary node **110** will be described next with reference to FIGS. **13A-13K** and **14**.

[0120] FIG. **13A** shows an example of secondary node transmitter **304** in greater detail. Transmitter **304** includes a plurality of circuits or switches SW. In this example, nine switches (SW-**0** to SW-**8**) are shown, although more or fewer switches may be provided than that shown in FIG. **13A**. Each switch may have, in one example, two inputs: the first input may receive user data, and the second input may receive control information or signals (CNT). Each switch SW-**0** to SW-**8** further receives a respective one of control signals SWC-**0** to SWC-**8** output from a control circuit. Based on the received control signal, each switch SW-**0** to SW-**8** selectively outputs one of a respective one of data streams D-**0** to D-**8**, a respective one of control signals CNT-**0** to CNT-**8**, to DSP **1302**.

[0121] Alternatively, in a manner similar to that noted above, the transmitted subcarrier can be removed by inserting zero amplitude at the input or output of the FFT, or by programming zero coefficients into selected multiplier inputs of pulse shape filters **1008** and **1016**, if a user desires that particular subcarrier is to be omitted such that fewer subcarriers are to be transmitted.

[0122] DSP **1302** may have a similar structure as DSP **902** described above with reference to FIGS. **9** and **10A**. In some instances, however, DSP **1302** may have a lower capacity than DSP **902** and, therefore, the number of circuits, such as FEC encoders, bits-to-symbol mappers, overlap and save buffers, FFT circuits, replicator circuits, and pulse shape filters may be reduced, for example, in accordance with the number of inputs to DSP **1302**. Accordingly, fewer subcarriers may be output from each of secondary nodes **112** compared to the number of subcarriers output from primary node **110**.

[0123] Based on the outputs of switches SW-**0** to SW-**8**, DSP **1302** may supply a plurality of outputs to D/A and optics block **1301**, which may have a similar construction as D/A and optics block **901** described above to supply X and Y polarized optical signals, each including I and Q components, that are combined by a PBC and output onto an optical fiber segment **1316** included in one of optical communication paths **115** shown in FIG. **2**.

[0124] Alternatively, based on zeroes (0s) stored or generated in DSP **1302**, subcarriers may be blocked or added in a manner similar to that described above.

[0125] FIGS. **13B** to **13E** show examples of subcarriers (SC') output from each of secondary nodes

112-j to **112-m**, respectively, in an upstream direction when, for example, subcarriers **SC0-SC19** shown in FIG. **6** are transmitted in the downstream direction. Namely, FIG. **13B** shows a first group of subcarriers **SC0'** to **SC4'** which may be output from secondary node **112-j**; FIG. **13C** shows a second group of subcarriers **SC5'** to **SC9'** which may be output from secondary node **112-k**; FIG. **13D** shows a third group of subcarriers **SC10'** to **SC14'** which may be output from secondary node **112-l**; and FIG. **13E** shows a fourth group of subcarriers **SC15'** to **SC19'** which may be output from secondary node **112-m**. As shown in FIG. **2**, each group subcarriers may be combined in combiner **116**, for example and supplied over optical communication path **117** to a receiver (e.g., receiver **204** shown in FIG. **3**) of primary node **110**. FIG. **13F** shows such combined optical subcarriers **SC0'** to **SC19'**, each having a corresponding one of frequencies **f0** to **f19**.

[0126] FIGS. **13G** to **13J** show example of subcarriers (**SC''**) output from each of secondary nodes **112-j** to **112-m**, respectively, in an upstream direction when, for example, subcarriers **SC0-SC19** shown in FIG. **8** are transmitted in the downstream direction. In particular, FIG. **13G** shows a first group of subcarriers **SC0''** to **SC6''** and **SC8''** which may be output from secondary node **112-j**; FIG. **13H** shows a second group of subcarriers **SC7''** and **SC9''-SC11''** which may be output from secondary node **112-k**; FIG. **13I** shows a third group of subcarriers **SC12''** to **SC16''** which may be output from secondary node **112-l**; and FIG. **13J** shows a fourth group of subcarriers **SC17''** to **SC19''** which may be output from secondary node **112-m**. As noted, each group subcarriers output from the secondary nodes may be combined in combiner **116**, for example and supplied over optical communication path **117** to receiver **204** in primary node **110**. FIG. **13K** shows such combined optical subcarriers **SC0''** to **SC19''**, each having a corresponding one of frequencies **f0** to **f19**.

[0127] FIG. **14** shows an example of primary node receiver **204** in greater detail. Primary node receiver **204** includes an Rx optics and A/D block **1400** having a similar construction as Rx optics and receiver block **1100** described above in connection with FIG. **11A**. For example, block **1400** may include a polarization beam splitter that received an upstream polarization multiplexed optical signal including optical subcarriers **SC0'** to **SC19'** or **SC0''** to **SC19''** supplied by optical fiber link **1401**, which may be, for example, an optical fiber segment as part of one of optical communication paths **115-k** to **115-m** noted above, a local oscillator laser, optical hybrids, photodiodes, capacitive coupler, TIA/AGC circuits and analog-to-digital (A/D) converter circuits similar to those discussed above and configured in a manner similar to that described above. Accordingly, block **1400** provides outputs **XI**, **XQ**, **YI**, and **YQ** similar to corresponding output supplied from block **1100**. The bandwidth associated with each of the components and circuits in block **1400**, however, may have a greater bandwidth than the corresponding components in secondary nodes **112**, and, therefore, primary node receiver **204** may detect and process each of the subcarriers output from secondary nodes **112**.

[0128] As further shown in FIG. **14**, primary node receiver **204** also may include DSP **1450** that receives signals **XI**, **XQ**, **YI**, and **YQ** from block **1400**. DSP **1450** may have a similar structure as DSP **1150** shown in FIG. **11A**. DSP **1450**, however, may have a higher capacity than DSP **1150** and, therefore, the number of circuits, such as FEC decoders, symbol to bit mappers, overlap and save buffers, CDEQ circuits, and PMDEQ circuits, may be greater than that shown in FIG. **11A** in accordance with the number of outputs from DSP **1450**.

[0129] The outputs from DSP **1450** next may be supplied to switches **SW-0** to **SW-19**, which selectively output, under the control of control signals similar to those described above, one of: a corresponding one of data streams **D-0** to **D-19**; and a corresponding one of control signals **CNT-0** to **CNT-19**. The control information may include, for example, monitoring information associated with secondary nodes **112** or scheduling information, such as time of flight (TOF) or delta time TOF information as described below with reference to FIGS. **20-24**.

[0130] Alternatively, based on zeros generated or stored in DSP **1450**, data may be selectively be supplied from each DSP output in a manner similar to that described above.

[0131] In a further example consistent with the present disclosure, guard bands or frequency gaps may be provided between adjacent subcarriers SC. As shown in FIG. 15, guard band GB1 may be provided between subcarriers SC4 and SC5, and guard band GB2 may be provided between subcarriers SC5 and SC6. Additional guard bands may be provided between remaining adjacent pairs of subcarriers. Such guard bands may be provided in order to detect and process each subcarrier more accurately by reducing crosstalk or other interference between the subcarriers.

[0132] As noted above, network configurations other than the aggregation network configuration discussed above in connection with FIGS. 1 and 2 are contemplated here, such as a mesh network, an example of which (network 1800) is shown in FIG. 16. Network 1800 may include a plurality of nodes 1802, 1804, 1806, 1808, 1810, and 1812, which are connected to have multiple paths between one another. For example, node 1802 may communicate with node 1812 via a path including nodes 1806 and 1808 or a path including nodes 1804 and 1810. Nodes 1802, 1804, 1806, 1808, 1810, and 1812 may include at least one primary node, similar to node 110 and at least one secondary node 112 similar to those described above. Further, in a manner similar to that described above the amount of data or data rate of data output from each of the nodes shown in FIG. 16 may be controlled or dynamically adapted based on control information exchanged between each node.

[0133] Allocation of data carried by subcarriers SC0 to SC19 and output from secondary nodes 112-j to 112-m, and examples of reallocation of such data output from the secondary nodes will next be described with reference to FIGS. 19 and 20a to 20c. FIG. 19 is a timing diagram showing transmission of control information on dedicated subcarriers SC2, SC7, SC12, and SC17, as further depicted in FIG. 4. As shown in the example depicted in FIG. 17, each of these subcarriers is continuously transmitted from primary node 110, while remaining subcarriers SC1, SC3-SC6, SC8-SC11, SC13-SC16, SC18, and SC19 are not transmitted. The control information carried by subcarriers SC2, SC7, SC12, and SC17 is associated with nodes 112-j, 112-k, 112-l, and 112-m, respectively. Moreover, subcarriers SC2, SC7, SC12, and SC17 can also carry user data associated with each such node, in another example.

[0134] FIG. 18A shows a timing diagram corresponding to the subcarrier transmission shown in FIG. 6. Here, data associated with node 112-j is continuously transmitted on subcarrier SC0 to SC4, data associated with node 112-k is continuously transmitted on subcarriers SC5 to SC9, data associated with node 112-l is continuously transmitted on subcarriers SC10 to SC14, and data associated with node 112-m is continuously transmitted on subcarriers SC15 to SC19.

[0135] As noted above, data associated with secondary nodes 112 may be reallocated among subcarriers SC0 to SC19. Consistent with present disclosure, such reallocation may occur on a time slot-by-time slot basis. An example of such reallocation is shown in FIG. 18B. Namely, as shown in FIG. 18B, during time slot TS1 subcarrier SC8 carries data associated with node 112-k. During time slot TS2, however, subcarrier SC8 carries data associated with node 112-j and continues to do so until time slot TS6. At which point subcarrier SC8 carries data associated with node 112-k and does so through time slot TS9, as well. In a similar fashion other time slots may be switched to carry data associated with different nodes, or, as further shown in FIG. 18B, subcarriers, such as subcarriers SC0 to SC4 may continue to carry data associated with just node 112-j. Such reallocation may be based on capacity requirements in the system. In addition, as noted above, the data associated with a particular secondary node 112 is that data which is output from the secondary node based on control information received by the second node and the configuration of the second node, as described above with reference to FIGS. 10B and 10C.

[0136] FIG. 18C shows a further example in which, in the absence of a particular capacity need during a given time slot, no data is transmitted. Namely, as shown in FIG. 18C, during time slot TS5 subcarriers SC11, SC8, and SC4 do not carry any. These subcarriers are blocked during time slot TS5 by circuitry described above with reference to FIGS. 10B and 10C.

[0137] The above examples describe reallocation of data to subcarriers transmitted in the downstream direction from primary node 110 to secondary nodes 112. In the upstream direction, as

described above, one or more subcarriers may be transmitted from one or more secondary nodes **112** to primary node **110**. If secondary nodes **112** are located at different distances away from primary node **110**, the arrival time of data at primary node **110** from one secondary node **112** may be different than the arrival time of data from another secondary node **112**. As a result, data input to the primary node **110** from secondary nodes **112** may not be synchronized, such that data launched on a given subcarrier from a first secondary node during one time slot may arrive while primary node **110** is receiving data from another secondary node, leading to errors or loss of data at primary node **110**.

[0138] For example, with reference to FIG. **19**, if node **112-l** is farther away from primary node **110** than node **112-m**, data output from node **112-l** on subcarrier SC4, for example, will experience a delay, td . Such data will not arrive at primary node **110** until time $t5+td$, as shown in FIG. **19**, and will continue to be received for the duration of a time slot, i.e., until time $t6+td$. At time $t6$, however, primary node **110** may begin to receive data from node **112-m**, and due to the delay data transmission from node **112-l**, primary node **110** will, in this example, receive data from both node **112-l** and **112-m** on subcarrier 4, such that neither data supplied from node **112-l** nor data supplied from node **112-m** may be reliably detected during the interval $t6$ to $t6+td$.

[0139] According to an aspect of the present disclosure, a method is provided by which circuitry in primary node **110** may determine the “time of flight” delay or the amount of time required for the primary node to send a request to a secondary node and for the primary node to receive a response from the secondary node. Based on such time of flight information, primary node **110** may then schedule or coordinate transmission on various time slots to avoid collisions, such as that shown in FIG. **19**.

[0140] FIG. **20** shows a flow chart of a method for determining time of flight information for each second node **112**. In step **2002**, circuitry in primary node **110** initializes the time of flight of flight (TOF) offset for each leaf **112** to zero. In step **2004**, a test message or TOF offset is sent to each leaf node **112** with an instruction or request for each such leaf node **112** to respond on a control channel, such as an in-band channel that is transmitted with data and is carried by one of the subcarriers. Circuitry in primary node **110** next measures a response time for each leaf node **112** (step **2006**). Circuitry in primary node **110** next determines the difference in TOF times (“delta TOF”) for each node and each delta TOF is adjusted so that the leaf node **112** having the longest TOF has a delta TOF of zero (step **2008**). Next, circuitry in primary node **110** determines when the delta TOF response time of each leaf node **112** is within a given tolerance or margin (step **2010**). If not, steps **2004**, **2006**, **2008**, and **2010** are repeated and will continue to be repeated until the delta response time is within such margin. If the delta response time is within the margin, the above method starts again at step **2002** to reassess TOF information after a predetermined time period of X milliseconds.

[0141] Once the TOF and delta TOF information is obtained by the method shown in FIG. **20**, primary node **110** schedules time slots to avoid collisions through a further method having an associated flow chart shown in FIG. **21**. In a first step **2102** of the scheduling method, hub **110** evaluates or measures by way of the control channel noted above leaf node utilization of subcarriers and time slots. Based on such measured utilization, circuitry in hub **110** determines whether a leaf node is within the TOF tolerance or margin (step **2104**). If yes, one or more time slots are allocated to that leaf node (step **2106**), and, if not, the hub node provides instructions so that one or more subcarriers are allocated to such leaf node such that such leaf node may continuously transmit data on such allocated subcarrier(s) to hub node **110**. The leaf nodes are then configured to transmit only during designated time slots or continuously over a one or more subcarriers in accordance with instructions received, for example, by way of the subcarriers noted above dedicated for sending control channel information (step **2110**). In step **2112**, the hub monitor leaf node utilization over a predetermined time period, such as a time period of Y milliseconds. The process next returns to step **2102**.

[0142] Note that the above TOF-related calculations may be performed outside of primary node **110** or by circuitry inside primary node **110**, as noted above.

[0143] FIG. **22** is a plot showing examples of time hub request and leaf response times, as well as corresponding TOF times. As depicted in FIG. **22**, the farther away a secondary node is from the primary node, the longer or the greater the TOF time. For example, secondary node **112-m** is located 100 km away from primary node **110** and is thus the farthest secondary node away from primary node **110**. Further, in this example, secondary node **112-l** is 60 km away from primary node **110**, and secondary node **112-k** is 50 km away. In addition, node **112-j** is located 30 km away from primary node **110** and is thus the closest secondary node to the primary node.

[0144] TOF values are determined by circuitry in primary node **110** that measures the time it takes for the primary node to receive a response from a secondary node following a TOF request response (see steps **2004** and **2006** above). As shown in FIGS. **22** and **23**, the TOF associated with second node **112-j** is 300 μ s (or 1.5 time slots (TS), assuming each time slot is 200 μ s); the TOF associated with node **112-k** is 500 μ s (or 2.5 TS); the TOF associated with node **112-l** is 600 μ s (3.0 TS); and the TOF associated with node **112-m** is 1000 μ s (or 5.0 TS). As noted above, after each TOF is determined, circuitry in primary node **110**, for example, determines the TOF associated with each node, wherein the delta TOF is the difference between the longest TOF, here 1000 μ s, and the TOF of each remaining secondary node **112** (i.e., longest TOF-TOF of each remaining secondary node). Accordingly, as further shown in table **2302**, the delta TOF associated with secondary node **112-j** is 700 μ s ($1000 \mu\text{s} - 300 \mu\text{s} = 700 \mu\text{s}$ or 3.5 TS); the delta TOF associated with secondary node **112-k** is 500 μ s ($1000 \mu\text{s} - 500 \mu\text{s} = 500 \mu\text{s}$ or 2.5 TS); the delta TOF associated with secondary node **112-l** is 400 μ s ($1000 \mu\text{s} - 600 \mu\text{s} = 400 \mu\text{s}$ or 2.0 TS); the delta TOF associated with secondary node **112-m** is 0.0 μ s ($1000 \mu\text{s} - 1000 \mu\text{s} = 0.0 \mu\text{s}$ or 0 TS).

[0145] In scheduling the transmission from secondary node **112** to primary node **110**, both TOF and delta TOF are taken into account in order to achieve desired arrival times at primary node **110** for each such transmission. An example of such scheduling will next be described with reference to FIG. **24**.

[0146] FIG. **24** shows example timing charts **2402** and **2404** in which, for ease of explanation, transmission on only subcarriers SC1 to SC4 is shown. Chart **2402** shows launch timing and leaf node utilization in absolute time for the secondary nodes have the TOFs and delta TOFs noted in FIGS. **22** and **23**. Chart **2404** shows data carried by various subcarriers originating at such secondary nodes and corresponding arrival times at the primary node without any collisions also in absolute time. In one example, as shown in FIG. **24**, it is desirable that node **110** receive data on subcarriers SC1, SC2, and SC4 from secondary nodes **112-j**, **112-k**, and **112-m** respectively, at the beginning of time slot TS6. As noted above, since secondary node **112-m** is the farthest from primary node **110**, secondary node **112-m** has an associated delta TOF of zero (0). Data is thus scheduled to be transmitted, as shown in chart **2402**, during time slot TS1, and, as shown in chart **2404**, such data arrives five time slots later during time slot TS6 (see FIGS. **22** and **23**). Node **112-k**, on the other hand, has a shorter time of flight of 2.5 TS. In order to arrive at node **110** at approximately the same time as the data output from node **112-m**, transmission from node **112-k** is scheduled to be output or launched toward node **110** following the launch of the node **112-m** data by an amount of time or delay D-k in chart **2402** approximately equal to delta TOF associated with node **112-k** or 2.5 TS (see chart **2402**). As a result, following such delay, the node **112-k** data is further delayed by the TOF associated (2.5 TS) with node **112-k**, such that the **112-k** data arrives at approximately the same time as the **112-m** data (see chart **2404**).

[0147] Further, in order for data transmitted from node **112-j** to arrive at node **110** at approximately the same time as data output from node **112-m**, transmission from node **112-j** is scheduled to be launched toward node **110** after the launch of the node **112-m** data by an amount of delay D-j approximately equal to the delta TOF associated with node **112-j** or 3.5 TS (see chart **2402**). Accordingly, following D-j, the node **112-k** data is further delayed by the TOF associated (1.5 TS)

with node **112-j**, such that the **112-j** data arrives at approximately the same time as the **112-m** data (see chart **2404**).

[0148] Other example launch and arrival times and subcarrier utilization are further shown in FIG. **24**.

[0149] In another example, empty time slots may be provided between transmission by one node on a given subcarrier and transmission by another node on that subcarrier in order to provide sufficient time between slots so as to reduce the risk of collisions or data transmission on a given subcarrier in which time slots overlap, for example, as shown in FIG. **19**.

[0150] Various modifications and other embodiments will be apparent to those skilled in the art from consideration of the present specification, and the detailed implementations described above are provided as examples. For example, the digital signal process disclosed above may be implemented as a programmable gate array circuit (PGA), or a field programmable gate array circuit (FPGA). In addition, although separate lasers **908** and **1110** are provided in the transmitter and receiver, respectively, as noted above, a transceiver consistent with the present disclosure may include a common laser that is “shared” between the transmitter and receiver. For example, FIG. **25** is a diagram illustrating an example of the transceiver **110** using a shared laser **2502** providing optical signals both for transmission and reception (as a local oscillator signal) in accordance with one or more implementations of the present disclosure. As shown, the laser **2502** generates an optical signal and provides the optical signal to the splitter **2504**. The splitter **2504** splits the optical signal into two portions. One portion is provided to the optical hybrids or mixers **1120-1** and **1120-2**, while the other portion is provided to modulators **910-1** to **910-4**.

[0151] Accordingly, other implementations are within the scope of the claims.

Claims

1. A method, comprising: transmitting a plurality of optical subcarriers from a first node to a second node in a network; transmitting first control information from the first node to the second node; receiving the plurality of subcarriers and the first control information at the second node; outputting from the second node first data indicative of first information carried by a first group of the plurality of optical subcarriers based on the first control information; transmitting second control information from the first node to the second node; and outputting second data from the second node indicative of second information carried by a second group of the plurality of optical subcarriers based on the second control information.
